

Detecting Keratoconus using Machine Learning Models

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Abstract — One of the major progressing sectors due to the introduction of technology has been healthcare. Diagnosis of patients has improved by manifolds. Keratoconus is a rare disease where it affects the patient's cornea. There are ongoing researches around the world to find a solution that is accessible and practical. Our objective is to detect whether a person is suffering from Keratoconus or not. This huge volume of important data cannot be handled manually, hence use of concepts like machine learning, data analysis, data mining, etc. play an important role. To evaluate accuracy of Machine learning models like Inception V3, VGG16, MobileNet V2, ResNet 50 using color coded corneal maps. The authors have implemented these models and compared their performances amongst each other and thus select the best fit model. The training set contains of 1050 images and comprising of 1051 Normal eyes and 862 Suspect eyes. The models were implemented in python language on the Google Colab Platform. These models are providing a range of 75-95% accuracies depending on the different models. The highest accuracy was obtained by Inception V3 which was 95%. The dataset were corneal maps recorded using Scheimpflug imaging system. Based on the classification of the parameters of the corneal maps, the input data was sorted on the basis of severity and also predicting how likely the patient is to suffer from keratoconus

Keywords— Machine Learning, Healthcare, Keratoconus, Deep Learning, Convolutional Neural Network, Corneal thickness

I. INTRODUCTION

Keratoconus is a rare corneal illness which takes on roughly every 45 people per 100,000 individuals. In this corneal disease, the cornea of the eyes tends to suffer from protrusion and bulging outwards. The shape is deformed and has tendency of forming a conical shape. It also causes the stroma to become thinner. Thus, resulting in weakened eyesight. Most advanced cases of keratoconus can be detected easily as the symptoms can be clear and visible. However, detecting early stages and suspect cases is where it becomes comparatively difficult. This requires more sensitive and comprehensive approach of assessment are various techniques to do so. Since, puberty and early adult life is where most keratoconus cases start developing from, it is very crucial to detect it at an early stage and take necessary remedy strategy. Treatments can include medications, certain practices prescribed by ophthalmologist, or even surgeries, if needed.

Developing predictive or detection models isn't an easy task as it has to work accurately on numerous types of eye data set. The model has to be efficient, accurate, and sensitive as well as has to have variability in corneal parameters that will be determining factor. Detecting suspect cases is even more difficult as there won't be major developments in symptoms. The corneal parameters that will

be involved in the assessment procedure can be of various types. The difference in corneal topography measurements plays a major role in diagnosis. Now, the corneal measurements can be based on various types of topography like, OCT topography, corneal topography which is based on Placido's discs, elevation-based topography, and generated elevations as well as topography based on imaging technology by Scheimpflug.

Our prediction/classification model takes the patients' corneal measurements on whichever topography as data. This data is processed on a machine learning module using various models or even combinations of some. In figuring, information is converted into a formation which is efficient in data handling and development. In comparison to the current transmission and PCC media, data will be converted into paired advance formations.

After the implementation of learning system, results are visualized and plotted in many possible ways. The objective of this project work is to review the already done work in the detection of keratoconus using various deep learning, machine learning algorithms. A summary of around 25 different methodologies have been summarized below in the tabular form for better understanding of the research work.

II. LITERATURE REVIEW

This research paper by [1] sums up the construction of an input-based classifier where the classification model can detect if a person has Keratoconus disease or not based on corneal topographic and tomographic parameters. Keratoconus is a rare eye disease where the patient's cornea is affected. The cornea starts bulging outwards hence impacting on vision. Early detection of the disease can provide to be helpful in treatment. The design of the classification model is retrospective case series. Corneal Measurements were taken by Scheimpflug camera along with a combination of Placido corneal topography. A total of 3502 eyes were used as dataset. Among them 877 eyes were of Keratoconus patients, 940 with history of corneal surgeries, 426 cases that associated to subclinical keratoconus and the rest were healthy eyes. The accuracy for cases of normal eyes was averaging 97%, for cases of keratoconus it was 95%, for cases of subclinical keratoconus it was 92% and for cases with history of corneal surgeries it was around 96%. These numbers for accuracy clearly show the classification algorithm have very good precision and detailed specificity in differentiating among categories. However, the key point to be noted was that inclusion of posterior parameters significantly improved the efficiency and precision. Some other limitations were that it wasn't very useful in cases with early signs of corneal ectasia. This was in conclusion, a classification model used to identify

cases of keratoconus, subclinical keratoconus and healthy eyes. No prediction feature was provided in this model.

The main objective of this report [2] was to evaluate comparisons between anterior corneal anomaly and topographic measurements in subjects suffering from keratoconus and non – keratoconus (healthy) eyes. The dataset collected were taken by Scheimpflug – Placido Topography method. A total of 161 cases of eyes were analyzed and after analysis were categorized into three classes of Keratoconus based on their severity (i.e., Grade 1,2 & 3) and the rest were under healthy eyes. 86 of them were male and 75 were female amongst them. Amsler-Krumeich Classification method was used to carry out the analysis process. The major factors that played the most important role in the classification process were mean measurements, central corneal thickness, and aberrations. After all categorization of cases into their respective groups, some inferences of comparisons were drawn like further classification based on gender, age, etc. The algorithm has very good efficiency and accuracy. There wasn't much difference on basis of age and gender distribution.

The study [3] used different machine learning models to analyses Keratoconus dataset. The dataset contains 3151 corneal images and ESI (Ectasia Screening index) was used to determine the grade of keratoconus. ESI [from 0 to 4] means normal, [between 5 to 29] means suspect and over 30 means keratoconus disease. Pre-processing of data to remove unnecessary parameters. The greedy hill climbing algo determines predictive ability of the data.

The aim of this system [4] built, was to check precision and accuracy of a scoring system designed to differentiate keratoconus eyes and healthy eyes with the help of a Scheimpflug camera system tomography. 39 parameters under Scheimpflug imaging system were used for the evaluation. The dataset consisted of 512 non-keratoconus eyes and 624 cases of keratoconus eyes. The summation of all the scores gave us the absolute score. 32 out of the 39 parameters showed at most 1 score in non-keratoconus subjects. TKS showed very good predictive accuracy for diagnosis with 95% accuracy. It also showed very good sensitivity and specificity percentages(both around 99%) at -12.45 being the best cut-off point. Use of pachymetry data and topometric indices can improve the accuracy even more. The major outputs of the analysis were sensitivity, specificity, accuracy, and cutoff.

The report by [5] carried out detailed research. Keratoconus is a rare eye disease where the patient suffers visual loss due to irregularity in corneal astigmatism. Treatment includes interventions or even surgery for severe cases. The report contains the methodical survey and meta-investigation of 11529 eyes comprising of healthy and keratoconus and other categories. The ratio of systematic review and meta-analysis was 41:23 respectively and the overall number of publications being searched amounted up to 4000 titles. There was a combination of analysis and prediction model. Regression model used for prediction found that the case subjects depicted 0.8 Diopters lower Maximum keratometry over a year of every increase in 10-year age. Mixed effects meta- regression was implemented for analyzing the influence of factor variables like age, topographic protocol, etc. The key highlighting conclusions made from the study helped in giving an idea on which age group or conditions might be more vulnerable to the disease of keratoconus. Patients of younger age and those with mean

keratometry more than 55 Diopter, are the most likely group to be a victim of progression of Keratoconus. Hence, they are mostly advised to consult with doctors at early stage and if needed may adopt low threshold cross- linking to avoid extreme case of keratoconus.

The report by [6] summarizes the implementation and working of an unsupervised machine algorithm used to detect severity of Keratoconus disease in a patient and monitor it. Around 3,156 cases with ESI (Ectasia Status Index) score ranging from 0% to 100% were involved in the analysis.

The purpose of the report by [7] is to implement filtering of infancy keratoconus using parameters of keratometer. The plan of the calculation depends on assessment. At 5 critical centers in Japan, 123 eyes of same number of patients were enrolled with Amsler-Krumeich request stage 1 (subjects were at most 50 years old where mean age was 26.36 years); the ratio of male and female was 84:39 and 205 eyes of equal number of sound cases (with typical age of 26.20 years, and sex ratio of 139:66 in the respective order of male and female). The area under the gatherer working brand name curve of keratoconus document was 0.90. This was basically much improved contrasted with individual limits ($P < .001$). The mindfulness and expresses regards in the bundle were 85.0% and 86.7%, independently. Though the responsiveness/unequivocally was not efficient, the advanced assumption condition using auto-keratometer-decided limits allowed better isolation of early stages of the disease of keratoconus.

Study carried out by [8]. The purpose was to present another methodology for keratoconus location dependent on corneal topography saw in vivo got from a solitary Scheimpflug picture. The philosophy carried out was in the accompanying manner. Scheimpflug single-picture previews from 25 subjects of healthy eyes and 25 eyes suffering from disease of keratoconus were dissected; from every bunch, 5 sets were arbitrarily chosen to give out-of-test information.

III. DATASET

The training set contains of 1050 images and comprising of 1051 Normal eyes and 862 Suspect eyes [9]. To detect keratoconus, corneal maps are used. Every corneal map is represented by different colors. The variety ranges from cool colors (like blue, purple), to warm colors (like red, orange) and neutrals (like green). The Pentacam images mainly depicts four parameters, they are:

Axial maps

- Helps in assessment of irregularity in astigmatism.
- warm colors represent steep
- cool colors represent flat

Corneal Thickness

- This depicts how corneal thickness is distributed over a measured area.
- also known as pachymetry
- warm colors depict thinner it is
- cool colors represent more thicker

Anterior Elevation

- helps in assessment of regularity and location of astigmatism
- warm colors refer to how high the cornea is from fit sphere
- Cooler colors refer to how below the cornea is from best fit sphere.

Posterior Elevation

- this helps in identification of former frustekeratoconus
- Warm colors and cooler colors represent the same as in anterior elevation.

The corneal images are collected using Pentacam instruments. The color scales were based on decimal scale grading using microns for the mentioned parameters. These parameters help in not only identifying keratoconus disease but also classify on the basis of severity. There are majorly 4 stages of keratoconus namely,

- Normal
- Suspect,
- Mild,
- Severe.

The ranges of the corneal parameters help in classification of their stages. The training set contains of 1050 images and comprising of 1051 Normal eyes and 862 Suspect eyes. The following table represents the values of parameters on the basis of which keratoconus and normal eyes are classified. Original Image is shown in Fig 1.

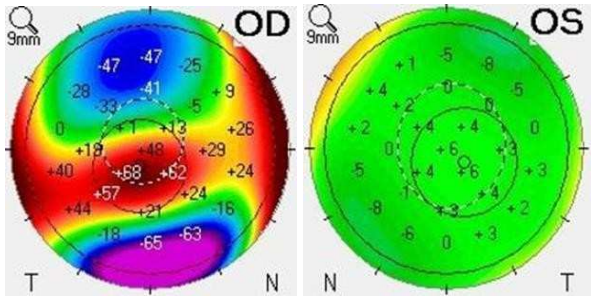


Fig1. Corneal images of Keratoconus VS Healthy Eyes

IV. METHODOLOGY

In this paper, the methodology or the working of the paper was categorized into 5 different steps. These steps included data acquisition, data pre-processing, model training, model testing and model evaluation. All these steps are described in detail below and the workflow of the methodology is shown in fig.2.

(i) Data Acquisition: This step involves collection of raw datasets for the analysis. Accumulation of data from relevant sources before proceeding towards storage, pre-processing is known as data acquisition.

(ii) Data pre-processing: The process of processing the raw data into usable data in order to prepare for prediction model developed. The quality of the data is evaluated in this step. Some characteristics on which quality is determined are accuracy, completeness, consistency, interpretability, etc.

(iii) Model Training: In this step, we instruct our machine to learn patterns and behaviors from our dataset allotted for training. Before this, we divide our data into training and testing sets.

(iv) Model Testing: After the training process, the algorithm will carry out testing to evaluate the performance of the training. On the basis of its training, the model shall be implemented on the testing dataset.

(v) Model Evaluation: In this step, we evaluate characteristics of our model like accuracy, precision and recall.



Fig 2. Methodology Workflow

In this paper we have taken the aid of deep learning models and image detection techniques to develop our model that analyses corneal parameters from the input data and then categorizing the results into the four said categories of keratoconus also predicting how vulnerable the patient is, to be suffering from the disease. Convolutional Neural Networks (CNN) [10] also played a very important role in our model. Neural Networks use RAM and GPUs which is why models built are more accurate and in a timely manner. Concept of transfer learning brings more advantages like extracting information like features, weight of layers, etc. from the models which have been trained. There is a variety of pre-trained models with different size, accuracy, depth and parameters. In this paper we have mainly used four Keras models namely ResNet50, VGG16, MobileNet V2, and InceptionV3. We implemented these models on our data and evaluated their performances in detecting Keratoconus.

V. MODELS ARCHITECTURE

Different machine learning models are used to predict keratoconus with the help of color-coded corneal maps. And in this section, we are going to describe elaborately.

A. VGG 16

VGG16 is a straightforward as well as generally utilized Convolutional Neural Network (CNN) Architecture utilized in ImageNet, huge visual data set venture utilized in visual item acknowledgment programming research. In ImageNet, the VGG16 model resulted with 92.7% accuracy with top-5 test precision, which is a dataset containing 14 million pictures along with 1000 classes. In 2004, it was one of the most efficient models submitted to ImageNet Large Scale Visual Recognition Challenge (ILSVRC). It generated upgrades over AlexNet design by providing massive amount of piece measured channels (first and second convolutional layer with 11 and 5) with multiple three $\times$ three-bit estimated channels consistently.

B. INCEPTION V3

On the dataset of ImageNet, inception V3 is the only image recognition model which resulted with the accuracy of more than 78%. This model is developed by combining

multiple researches works of various researchers who worked with ImageNet. This predictive model is generated using different asymmetric as well as symmetric blocks which includes features like max and average pooling, dropouts, convolutions, etc. In order to boost the output, this model uses batch normalization vigorously.

C. MOBILENET V2

MobileNetV2 is a convolutional network plan that performs on mobile phones. It relies upon a changed excess development where the waiting affiliations are between the bottleneck layers. The widely appealing expansion layer uses lightweight profundity-wise convolutions to channel features as a wellspring of non-linearity. Generally, the design of MobileNetV2 contains the basic totally convolution layer with 32 channels, followed by 19 extra bottleneck layers.

D. RESNET50

ResNet50 or Deep Residual Network engineering is the pre-prepared profound learning model which has around 25,636,712 boundaries. The size of the model of only 98 MB with top-5 exactness of 92.1%. This exactness alludes to the model's exhibition on the ImageNet approval dataset. The ResNet50 engineering is partitioned into four phases. The most widely recognized size during tests is viewed as 224 x 224 x 3. Here 3 alludes to the variety channels of RGB. Toward the start, there are 2 layers, initially, a convolution layer of 7x7 bit size and furthermore a maximum pooling layer of 3x3 part estimates.

VI. RESULTS AND ANALYSIS

This section will represent each of the model's executed results i.e., how accurate and precise each model is for our particular specified application. In this paper, the authors have plotted the results in a visualized manner and hence compared amongst each other. Visual plot is shown in Fig 3.

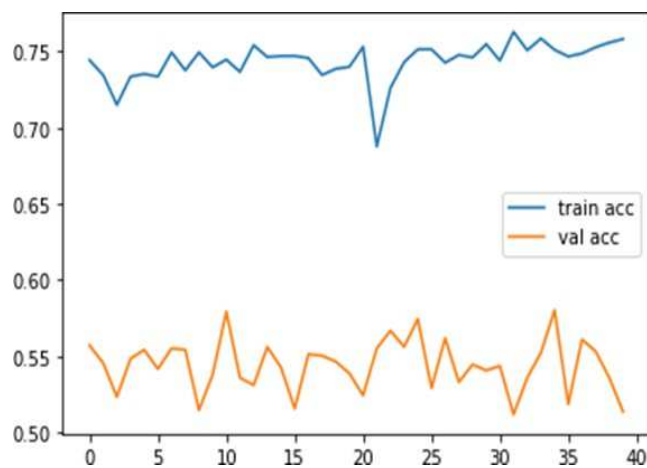


Fig 3. Visual Plot of accuracy in MobileNet V2

The overall accuracy of MobileNet V2 model was roughly around 72%. The major observation from the plots can be seen that the training accuracy is fairly decent (around the range of 75- 80%) which can obviously be improved majorly. But however, the anomaly here is the big difference between the training accuracy and validation accuracy. One of the major reasons behind this is the data being irregular and unorganized. Application of techniques like regularization, augmentation, etc. and combination of these

improved the difference slightly but there is a massive room for improvement in this concern. Visual plot is shown in Fig 4.

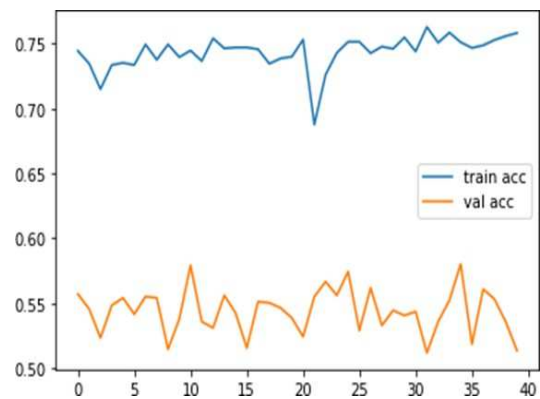


Fig 4. Visual Plot of accuracy in VGG 16

The results obtained was around 93% accuracy with the aid of VGG 16 model. Below are the visual plots of the values. The significant perception from the plots should be visible that the preparation precision is genuinely fair (around the scope of 90-95%) which can clearly be improved significantly. Yet, be that as it may, the significant issue here is the enormous distinction between the validation accuracy and training accuracy. One of the significant purposes for this is the information being unpredictable and heavily distorted. Utilization of methods like regularization, augmentation, and so forth and blend of these superior the distinction somewhat however there is an enormous opportunity to get better in this regard. Visual plot is shown in Fig 5.

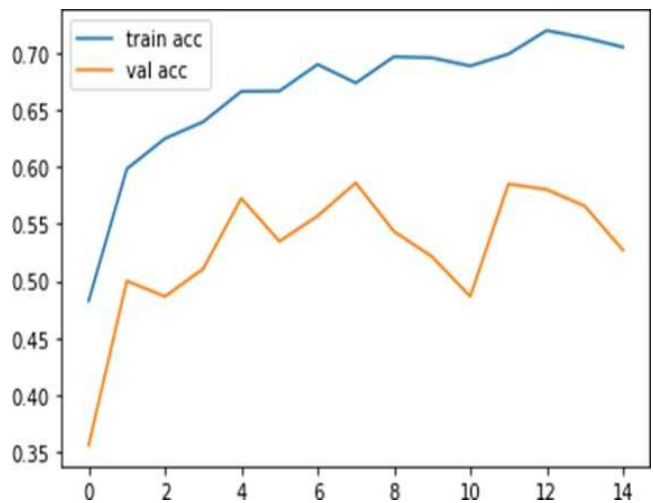


Fig 5. Visual Plot of Accuracy in ResNet 50

The results obtained was around 70.48% accuracy with the aid of ResNet50 model. Below are the visual plots of the values. The major observation from the plots can be seen that the training accuracy is fairly decent (around the range of 75- 80%) which can obviously be improved majorly. But however, the anomaly here is the big difference between the training accuracy and validation accuracy. One of the major reasons behind this is the data being irregular and unorganized. Application of techniques like regularization, augmentation, etc. and combination of these improved the difference slightly but there is a massive room for improvement. Visual plot is shown in Fig 6.

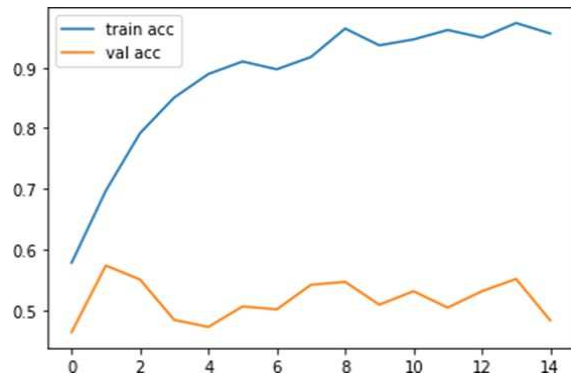


Fig 6. Visual Plot of accuracy in Inception V3

The results obtained was around 95% accuracy with the aid of Inception V3 model. Below are the visual plots of the values.

The major observation from the plots can be seen that the training accuracy is more than satisfactory which can obviously be improved majorly. But however, the anomaly here is the big difference between the training accuracy and validation accuracy.

One of the major reasons behind this is the data being irregular and unorganized. Application of techniques like regularization, augmentation, etc. and combination of these improved the difference slightly but there is a massive room for improvement in this concern. Models and accuracy achieved shown in Table I.

TABLE I. MODELS AND ACCURACY

Sl.No.	Models Used	Accuracy Achieved
1.	ResNet50	70.48%
2.	Inception V3	95%
3.	MobileNet V2	72%
4.	VGG16	93%

VII. CONCLUSION

Early detection of a disease like keratoconus is very crucial for its diagnosis. Earlier the detection, the easier will be the cure. Keratoconus is a rare eye disease that is most vulnerable for the teenage and adolescent stages which is considered to be a vital stage of life for development. The analysis of variety of models already implemented showed that a combination of anterior and posterior corneal topographic parameters gave better efficiency, accuracy and sensitivity. In the four models executed, the authors have used Keras models namely VGG16, ResNet 50, Inception V3 and MobileNet V2 on data collected from various sources like clinical centers, researchers, free source, etc. The dataset were corneal maps recorded using Scheimpflug imaging system. Based on the classification of the parameters of the corneal maps, the input data was sorted on the basis of severity and also predicting how likely the patient is to suffer from keratoconus. First using data pre-processing methodologies, data was cleaned and sorted. Then, by use of deep learning algorithms the main model was constructed and their performances were evaluated and compared. One of the biggest challenges during our project was collecting the dataset source. As it was clinical data, it was not openly available on the internet or archives. After multiple attempts of communicating to researchers and scholars who have previously worked with the topic. The Keras models were constructed in Python Language on the Google Colab platform. As discussed above, one of the major areas of improvement will be to bridge the gap between the validation

and training accuracy. One of the ways, it was figured is to have a larger sample size of data or to construct a more effective and detailed algorithm that processes the input data and makes it more workable. The current models are providing a range of 75-95% accuracies depending on the different models. But the major issue faced was the big difference between training and validation accuracies.

In previous papers, it was observed that various models like SVM (Support Vector Machines), Fuzzy C-Means Clustering, KNN, CNN being used to execute the same purpose. The existing models show fair precision, sensitivity and specificity but there are shortcomings in each of them. The broader objective was to construct a model that compensates for each of the shortcomings and also gave priority to introduce prediction that helps in predicting if a patient is likely to suffer from the disease or not.

ACKNOWLEDGEMENT

We would look like to express our gratitude towards Dr. Siamek Yousefi who guided us with resources.

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